

LECTURE NOTES ON EEE 435 (DIGITAL COMMUNICATION III)

CHAPTER ONE:

PULSE CODE MODULATION (PCM)

Definition: This is a system of pulse modulation in which an analog signal $x(t)$ is sampled and then the samples are quantized and encoded.

ADVANTAGES OF PCM OVER ANALOG MODULATION

1. In long distance communications, pcm signals can be completely regenerated at each repeater station if the repeater spacing is such that the magnitude of the noise is less than half the separation between levels.
2. At low input signal-to-noise ratios, the output signal-to-noise ratio of pcm systems is better than analog modulation schemes.
3. PCM systems can be designed to handle a variety of signals. A pcm system designed for analog message transmission can be readily adapted to handle other, particularly digital data.
4. Modulation and demodulation circuitry in pcm systems are all digital, thus affording high reliability and stability.
5. It is easy to store and time scale pcm signals. For example, pcm data gathered intermittently from an orbiting satellite over a period of hours may be transmitted to the ground station in a matter of seconds when the satellite appears in the field of view of the receiving antenna.
6. In a pcm system, source coding and channel coding techniques can be used to reduce unnecessary repetition (redundancy) in messages and to reduce the effects of noise and interference. Coding may also be used to make the digital communication channel more secure.
7. Exchange of bandwidth for power is easy to accomplish in pcm systems. In pcm systems can be easily time scaled, time can also be exchanged for signal power.

DISADVANTAGE

The main disadvantage of pcm system is its high complexity which is much greater than that required for other types of modulation systems.

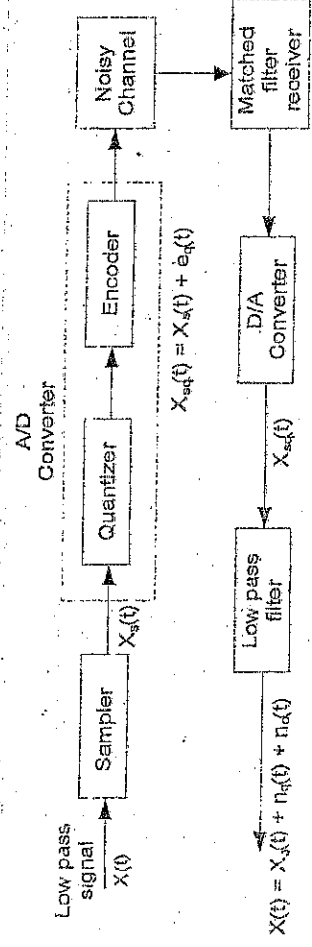


Figure 1: Block Diagram of a pcm system.

QUANTIZATION

This could be defined as the representation of sampled analog signal values by a finite set of signal amplitude levels. While sampling converts a continuous time signal to a discrete time signal, quantization converts a continuous amplitude sample to a discrete amplitude sample. Thus sampling and quantizing operations convert the output of an analog information source into a sequence of levels (or symbols), that is, analog source is transformed to a discrete (digital) source.

LINEAR QUANTIZATION

This is a method of quantization in which the range of continuous random variable X is divided into Q intervals of equal length, say Δ . If the value of X falls in the i^{th} quantizing interval, then the quantized value of X is taken to be the midpoint of the interval.

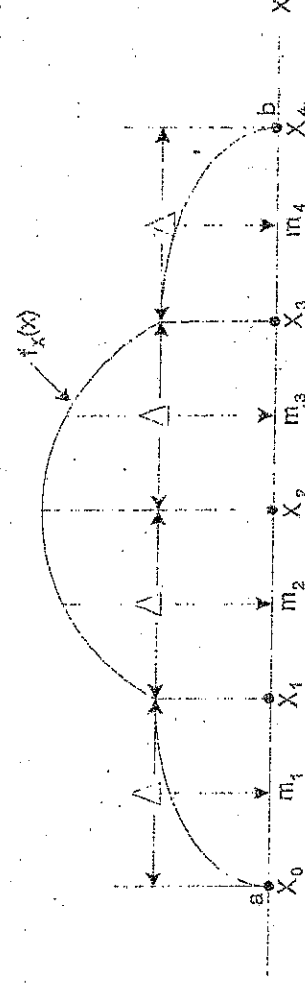


Figure 2: Linear Quantization, with Step size = Δ , $Q = 4$

If a and b are taken to be the minimum and maximum values of X , respectively, then the step size or interval length Δ is given by,

$$\Delta = (b - a)/Q \dots\dots\dots (1)$$

The quantized output X_q is generated according to,

$$X_q = m_i \text{ if } X_{i-1} < X \leq X_i \dots\dots\dots (2)$$

Where $X_i = a + i\Delta \dots\dots\dots (3)$

And $m_i = \frac{X_{i-1} + X_i}{2}, i = 1, 2, 3, \dots\dots\dots (4)$

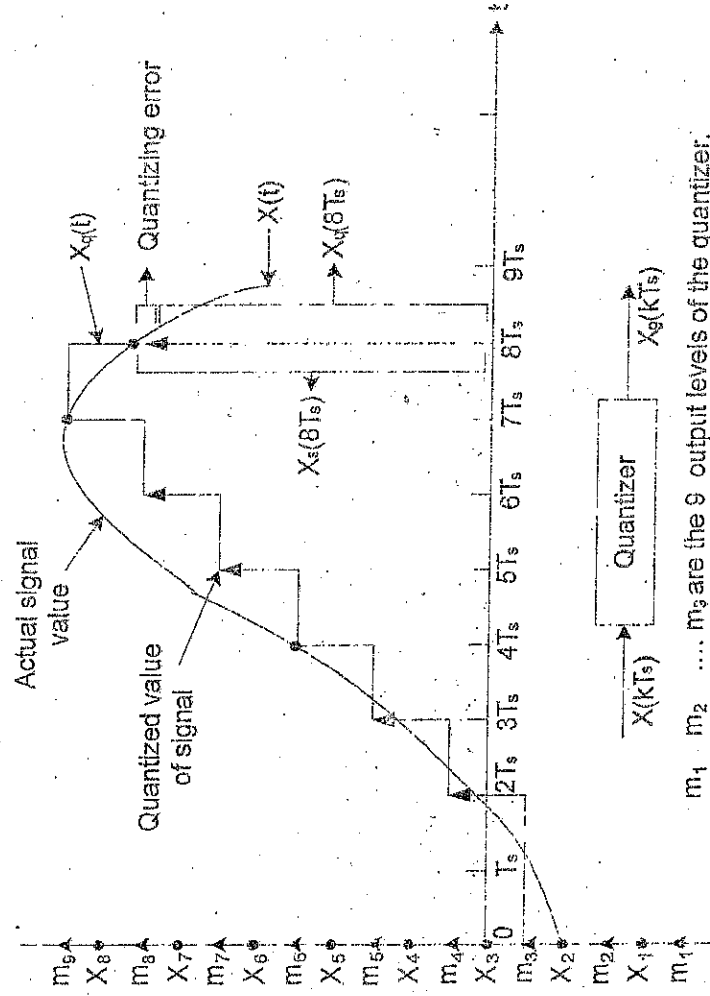


Figure 3: Input/Output Characteristics of a Linear Quantizer.

The output of the quantizer is a sequence of levels as shown in the figure as a waveform $X_q(t)$, where

$$X_q = X_q(kT_s), \quad kT_s \leq t < (k+1)T_s$$

Thus the quantized signal is a good approximation to the original signal. The quality of the approximation may be improved by a careful choice of x_i 's and m_i 's, such that some measure of performance may be optimized. The measure of performance that is most commonly used for evaluating the performance of a quantizing scheme is the output signal to noise power ratio (or signal to quantization error ratio).

The quantizer error (quantizing noise power) N_q for the uniform quantizer is given by,

$$N_q = E\{(X - X_q)^2\}$$

$$\begin{aligned}
 &= \int_a^b (x - x_q)^2 f_x(x) dx \\
 &= \sum_{i=1}^Q \int_{x_{i-1}}^{x_i} (x - m_i)^2 f_x(x) dx \dots\dots\dots 1
 \end{aligned}$$

Where $x_i = a + i\Delta$ and $m_i = a + i\Delta - \Delta/2$.

The signal power S_q at the output of the quantizer can be obtained from

$$S_q = E\{(X_q)^2\} = \sum_{i=1}^Q (m_i)^2 \int_{x_{i-1}}^{x_i} f_x(x) dx \dots\dots\dots 2$$

The ratio S_q/N_q gives a measure of fidelity of the linear quantizer. This ratio can be computed if the mean stationary random process of X (pdf of X) is known.

Example:

The input to a Q -step uniform quantizer has a uniform pdf over the interval $[-a, a]$. Calculate the average signal to quantizer noise power ratio at the output.

Solution: From equation (1)

$$\begin{aligned}
 N_q &= \sum_{i=1}^Q \int_{x_{i-1}}^{x_i} (x - m_i)^2 \left(\frac{1}{2a}\right) dx \\
 &= \sum_{i=1}^Q \int_{-a+(i-1)\Delta}^{-a+i\Delta} \left(x + a - i\Delta + \frac{\Delta}{2}\right)^2 \frac{1}{2a} dx
 \end{aligned}$$

Let $y = x + a - i\Delta + \frac{\Delta}{2}$ then $dy = dx$

$$\text{Hence } N_q = \sum_{i=1}^Q \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} y^2 \left(\frac{1}{2a}\right) dy = \sum_{i=1}^Q \frac{1}{2a} \left[\left(\frac{y}{3}\right)^3\right]_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}}$$

$$N_q = \sum_{i=1}^Q \left(\frac{1}{2a}\right) \left(\frac{\Delta^3}{12}\right) = \frac{Q\Delta^3}{(2a)12} = \frac{\Delta^2}{12} \quad \text{Since } Q\Delta = 2a.$$

Now the output signal power S_q can be obtained using equation (2)

$$S_q = E[(X_q)^2] = \sum_{i=1}^Q (m_i)^2 \frac{1}{2a} = \frac{Q^2 - 1}{12} (\Delta)^2$$

Thus the average signal to quantizer noise power ratio is,

$$\frac{S_q}{N_q} = \frac{Q^2 - 1}{Q^2} \approx Q^2 \quad \text{when } Q \gg 1$$

$$\therefore \left(\frac{S_q}{N_q}\right)_{dB} = 20 \log Q$$

NON-LINEAR QUANTIZATION

This is a method of quantization in which a range of continuous random variable X is divided into Q – intervals of unequal lengths (variable step sizes).

It has two important advantages over the linear quantization method.

- (i) Has a higher average signal to quantizing error ratio (or signal to quantizing noise power ratio) than the linear quantizer when the signal pdf is non – linear, which is the case in many practical situations.
- (ii) The rms value of the quantizing error of a non linear quantizer is substantially proportional to the (instantaneous) sampled value X and hence the effect of quantizer noise is masked (see diagram below).

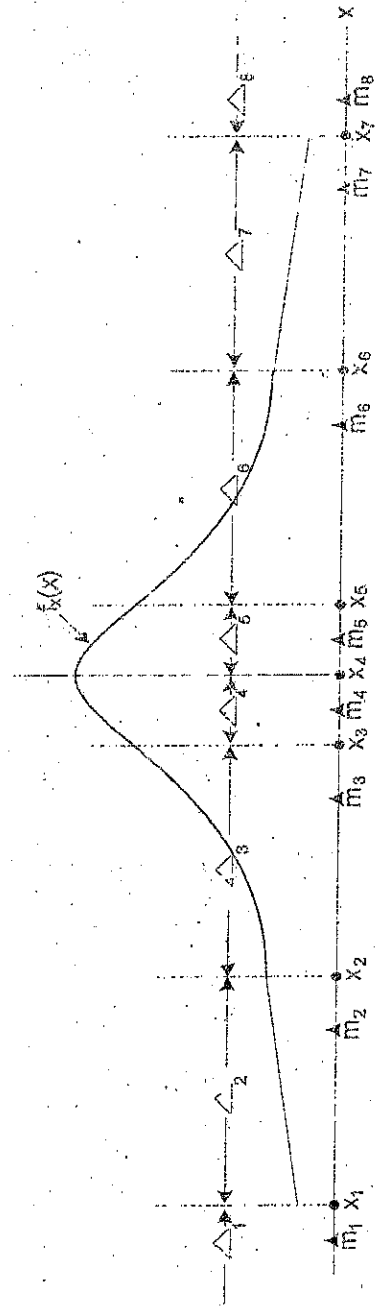


Figure 4: A non – linear quantizer for a Gaussian random variable.

$X_0 = -\infty$, $X_Q = \infty$, $Q = 8$, and $\Delta_i = \Delta_{Q-i+1}$ ($i = 1, 2, 3, 4$)

The input to the quantizer is a Gaussian random variable and the quantizer output is determined according to,

$$X_q = m_i \text{ if } X_{i-1} < X \leq X_i, \quad i = 1, 2, \dots, Q$$

$$X_0 = -\infty, \quad X_Q = \infty$$

The step size $\Delta_i = X_i - X_{i-1}$ is variable. The quantizer end points x_i and the output levels m_i are chosen to maximize the average signal to quantizing error ratio. In practice a non uniform (linear) quantizer is realized by sample compression followed by a uniform quantizer. Compression transforms the input variable X to another variable Y using a non – linear transformation;

$$Y = g(X);$$

Such that $f_Y(y)$ has a uniform pdf. Then Y is uniformly quantized and transmitted. At the receiver, a complementary expander with transfer characteristic g^{-1} restores the quantized values of X . The compressor and expander taken together constitute a Compander. The most commonly used compander uses a logarithmic compression,

$Y = \log X$, where the levels are crowded near the origin and spaced farther apart near the peak values of X .

DIFFERENTIAL QUANTIZING SCHEME.

It has already been noted that continuous random process can be adequately represented by a sequence of its sampled values $X(kTs)$ and that the individual samples $X(kTs)$ can be approximated by a set of quantized levels. In the quantizing techniques considered thus far, each sample in the sequence $X(kTs)$ was quantized independently of the value of the preceding sample. But in many practical situations, due to the statistical nature of the message signal $X(t)$ and due to over sampling, the sequence $X(kTs)$ will consist of samples that are correlated with each other. Differential quantizing schemes take into consideration the sample to sample correlation in the quantizing process. For a given number of levels per sample, differential quantizing schemes yield a lower value of quantizing noise power than direct quantizing schemes.

Example: The message signal $X(t)$ in a communication system is a zero mean stationary Gaussian random process, and it is sampled at a rate of 10,000 samples per second ($T_s = 0.1$ m-sec). The normalized auto correlation function $R_{xx}(\tau)/R_{xx}(0)$ has a value 0.8 when $\tau = 0.1$ m-sec. Two quantizing schemes being considered are;

- (a) A non-uniform quantizer with $Q = 32$ operating on each sample independently,
- (b) A differential quantizer with $Q = 32$ which operates on successive differences $[X(kTs) - X((k-1)Ts)]$. Assuming that the mean squared error due to quantizing a normal random variable with variance σ^2 is $2\sigma^2 Q^{-2}$, find the mean squared error of the two quantizing schemes given above.

Solution: (a) For the quantizer operating independently on each sample, the mean squared error is given by,

$$N_o = 2\sigma_x^2 Q^{-2} \approx (2)(10^{-3}) \sigma_x^2$$

(b) In the differential quantizing scheme, the variable being quantized is

$$Y = X(kTs) - X((k-1)Ts) \text{ and the variance of } Y \text{ is given by,}$$

$$\sigma_y^2 = \sigma_x^2(kTs) + \sigma_x^2((k-1)Ts) - 2E[X(kTs)X((k-1)Ts)]$$

$$= 2\sigma_x^2 \left(1 - \frac{R_{xx}(T_s)}{R_{xx}(0)}\right) = 0.4\sigma_x^2$$

Hence the mean squared error of the differential quantizing is given by

$$N_q = 2(0.4) \sigma_x^2 Q^{-2} = 0.8\sigma_x^2(Q^{-2}) \approx 0.8(10^{-3}) \sigma_x^2$$

(This is less than the error associated with direct quantizing scheme)

DIFFERENTIAL PCM SYSTEMS

These systems are particularly more efficient when the sampled message signal has high sample to sample correlation. For example in transmission of picture (video) information, appreciable portions of the signal describe background information containing very little tonal variations. In such situations, when pcm is used, the code-words describe the value of the average background level. If these tonal values do not change appreciably, then essentially repeated sample values are being transmitted. One way to improve the situation is to send only the digitally encoded differences between successive samples. Thus a picture that has been quantized to 256 levels (eight bits) may be transmitted with comparable fidelity using 4 – bit differential encoding. This reduces the transmission bandwidth by a factor of 2. PCM systems using differential quantizing schemes are known as differential pcm (DPCM) systems.

DELTA MODULATION SCHEME

A differential pcm system that is particularly simple to implement results when the difference signal is quantized into two levels. The output of the quantizer is represented by a single binary digit, which indicates the sign of the sample to sample difference. This pcm system is known as delta modulation (DM)

Delta modulation systems have the merit that the electronic circuitry required for modulation at the transmitter and demodulation at the receiver is substantially simpler than the corresponding hardware needed for PCM and DPCM systems.

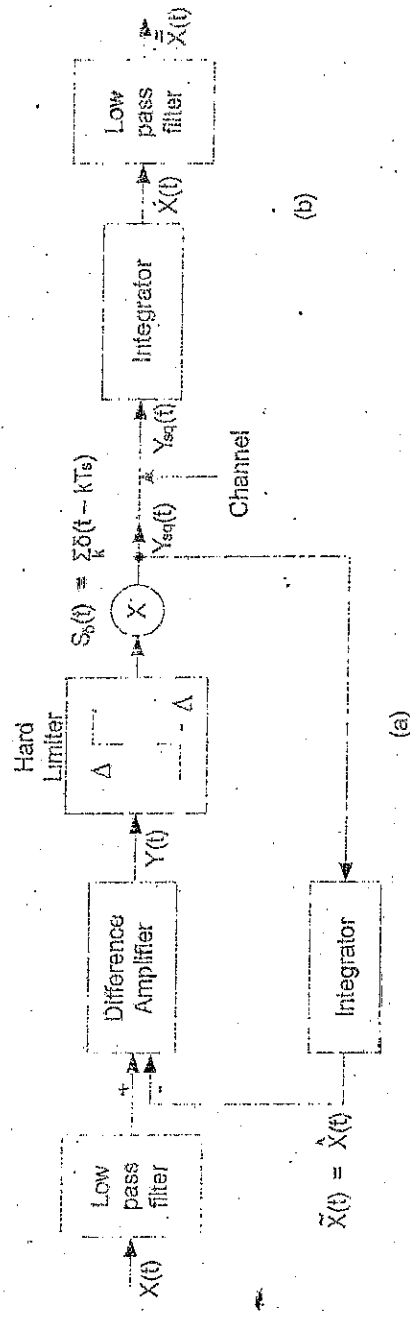


Figure 5: Block Diagram of DM System; (a) Modulator (b) Demodulator.

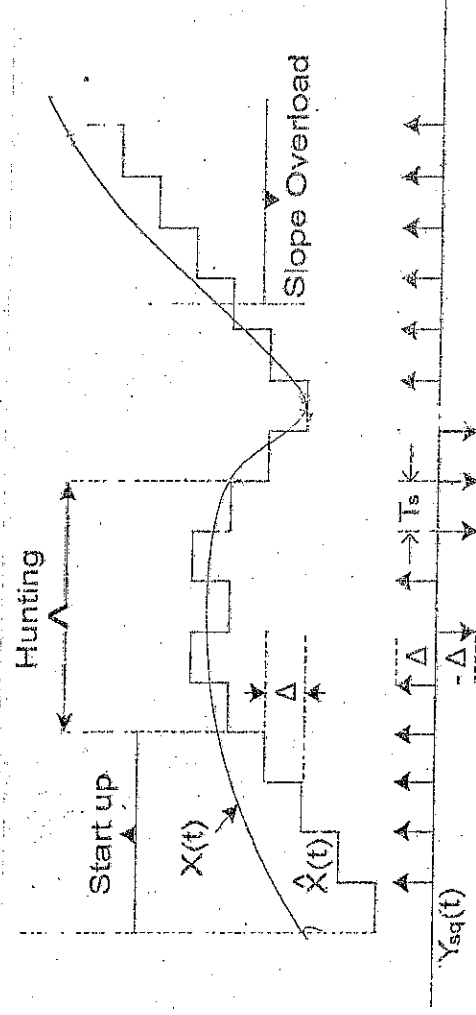


Figure 6: Delta Modulation Waveforms; $\hat{X}(t) = \tilde{X}(t)$

The operation of delta modulation system is shown in the block diagram in figure 5 above, and the operation of the system may be seen by using the waveforms shown in figure 6. The message signal $X(t)$ is compared with a stepwise approximation $\hat{X}(t)$ and the difference signal $Y(t) = X(t) - \hat{X}(t)$ is quantized into two levels $\pm\Delta$ depending on the sign of the difference. The output of the quantizer is sampled to

$$\text{produce, } Y_{sq}(t) = \sum_{k=-\infty}^{\infty} \Delta \text{sgn}[X(kT_s) - \hat{X}(kT_s)]\delta(t - kT_s) \dots\dots\dots (1)$$

The stepwise approximation $\hat{X}(t)$ is generated by passing the impulse waveform in equation (1) through an integrator that responds to an impulse with a step rise. Since there are only two possible impulse weights in $Y_{sq}(t)$, this signal can be transmitted using a binary waveform. The demodulator consists of an integrator and a low-pass filter.

In practical delta modulation system, the low-pass filter in the receiver will by itself, provide an approximate measure of integration; hence the receiver integrator can be eliminated to allow the filter to do the integration. At the transmitter, the sampling waveform $S_s(t)$ need not be an impulse waveform but a pulse waveform with a pulse duration that is short in comparison with the interval between pulses. Also the transmitter integrator need not be an ideal integrator. A simple RC low-pass filter will suffice. These simplifications reduce the complexity of the hardware in Delta Modulation systems considerably.

DIFFICULTIES ASSOCIATED WITH DM SCHEMES

Problems that occur with the use of delta modulation scheme to transmit analog signal can be deduced from the waveform shown in figure 6. Suppose at the starting period, $\hat{X}(t) < X(t)$ so that the first impulse has a weight of Δ . When this impulse is fed through the integrator, it produces a step change in $\hat{X}(t)$ of height Δ . This process continues until $\hat{X}(t)$ exceeds $X(t)$.

Hunting

During the start up interval, the receiver output will differ considerably from the message signal $X(t)$ until $\hat{X}(t)$ approach $X(t)$ and equalize. After the start up period (transient period) $\hat{X}(t)$ exhibit a hunting behavior when $X(t)$ does not vary (remains constant). Hunting leads to idling noise. The sampling rate in a delta modulation scheme will normally be much higher than the Nyquist rate and hence the rectangular idling noise waveform can be filtered or smoothed out by the receiver filter.

Slope Overloading

Another problem associated with the DM scheme is slope overloading. This overloading appears when the modulating signal changes between samplings by an amount greater than the step size Δ . Hence this type of overload is not determined by the amplitude of the modulating signal but rather by its slope – hence the name slope overloading.

PCM MULTIPLEXING (TDM/FDM)

TDM: Time Division Multiplexing, is a technique used for transmitting several analog message signals over a communication channel by dividing the time frame into slots and allocating one slot for each message signal.

FDM: Frequency Division Multiplexing is also a technique for transmitting several analog message signals over a communication channel by dividing the available bandwidth into slots, and allocating one slots for each message signal.

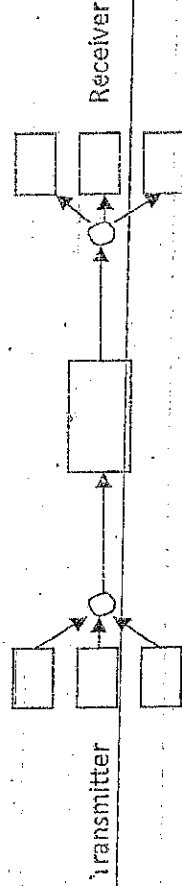


Figure 7: Block Diagramme of a 3-Channel TDM System

The diagramme illustrates the principle of TDM. The input signals, all band-limited to f_x by the input filters, are sequentially sampled at the transmitter by a rotary switch or commutator. The switch makes f_s revolutions per second and extracts one sample from each input during each revolution.

At the receiver, the samples from individual channels are separated and distributed by another rotary switch called a distributor or de-commutator. The samples from each channel are filtered to reproduce the original message signal. The rotary switches at the transmitter and receiver are carefully synchronized.

Example: To illustrate multiplexing with pcm, consider a case in which 1000 voice message signals and a single television signal are multiplexed. In this problem an 8-bit pcm is used for good fidelity, since 8-bits corresponds to $2^8 = 256$ quantization levels.

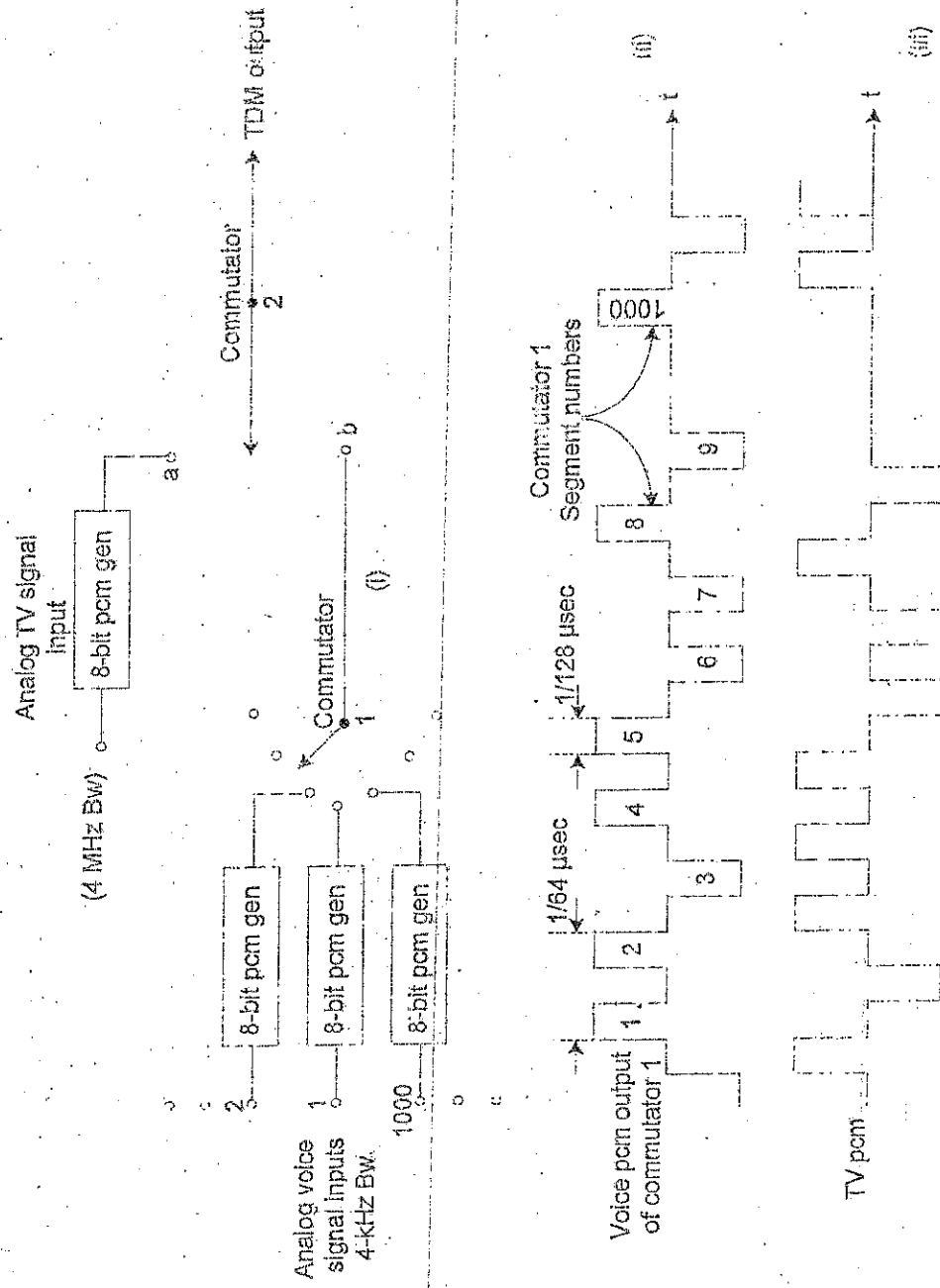


Figure 8: TDM of unequal bandwidth

A bandwidth of 4kHz is used for the voice message signals while for the TV signal a bandwidth of 4MHz is used. The multiplexing system shown in figure 8 uses two commutator switches.

For this problem, the commutator stays on a particular contact point only for the duration of one bit. The time between breaking contact at one switch point and making contact at the next is also equal to one bit. The output waveform of commutator 1 is as shown in figure 8 (ii). The pulse marked 1 is say, the first bit of 8 which will specify a sample value in channel 1. Pulse 2 is the first bit of 8 for a sample value in channel 2 etc.

Voice signals have a 4-kHz bandwidth, the sampling rate therefore is 8-kHz or sampling time is $1/8 \times 10^{-3}$ sec. Since there are 8-bits per sample, the commutator must make 8 revolutions in $1/8 \times 10^{-3}$ or 1 revolution in $1/64 \times 10^{-3}$ sec. Since there are 1000 voice signals, the time between bits in adjacent channels is $1/64 \times 10^{-6}$ sec ($1/64 \mu\text{-sec}$). The bit duration is $1/128 \mu\text{-sec}$ as shown in figure 8 (ii).

The Television signal with its 4-MHz bandwidth and 2-bit per sample generates a bit stream with bit separated by $\frac{1}{8(2 \times 4)} \times 10^{-6}$ sec or $\frac{1}{64} \mu\text{sec}$. The waveform of the TV bit stream is shown in figure 8 (iii). Note that all the operations must be done in synchronism. Commutator 2 must make contact with point b when commutator 1 is on a contact point. And when commutator 1 is between contacts, commutator 2 must make contact with point a. Similarly each pulse stream from each of the 1001 pcm generators must be timed so that individual bits fall in proper time slots without overlapping. Note that the product $4 \text{ kHz/channel} \times 1000 \text{ channels}$ yields 4 MHz which is the bandwidth of the television signal. Thus this relationship could allow easy interleaving of the voice and TV signals together. This is the reason why 1000 voice channels are used in the problem.

Exercises:

(1) Using block diagrams design a pcm multiplexing system to permit the combining of 3 signals, x_1 , x_2 and x_3 . The signals are band limited to 5 kHz, 10 kHz and 5 kHz respectively. Each signal is to be sampled at its Nyquist rate and quantized into 256 levels ($N = 8$). The commutator dwells on each signal so that the 8 bits are transmitted simultaneously.

(a) Sketch the system. What is the maximum bit duration that may be used?

(b) What is channel bandwidth required to pass the pcm signal.

(c) What is the commutator speed in revolutions per second?

(2) Three signals x_1 , x_2 and x_3 having bandwidths 5 kHz, 10 kHz and 20 kHz are each quantized into 256 levels. Each signal lasts for 10 minutes. If each signal is sampled at its Nyquist rate,

(a) How many samples are taken from each signal?

(b) How many bits are produced from each signal, if each signal is PCM encoded using 8 bits per sample?

CHAPTER TWO

TRANSMITTERS AND TRANSMITTER TESTING

Introduction: The function of a radio transmitter is to convert a source of electric power into an intelligence carrying radio frequency wave. The transmitter output power is fed to an antenna which releases the electromagnetic energy from the confines of a transmission line to an electromagnetic wave travelling through space.

Exercise: How is the energy released from the radiating antenna?

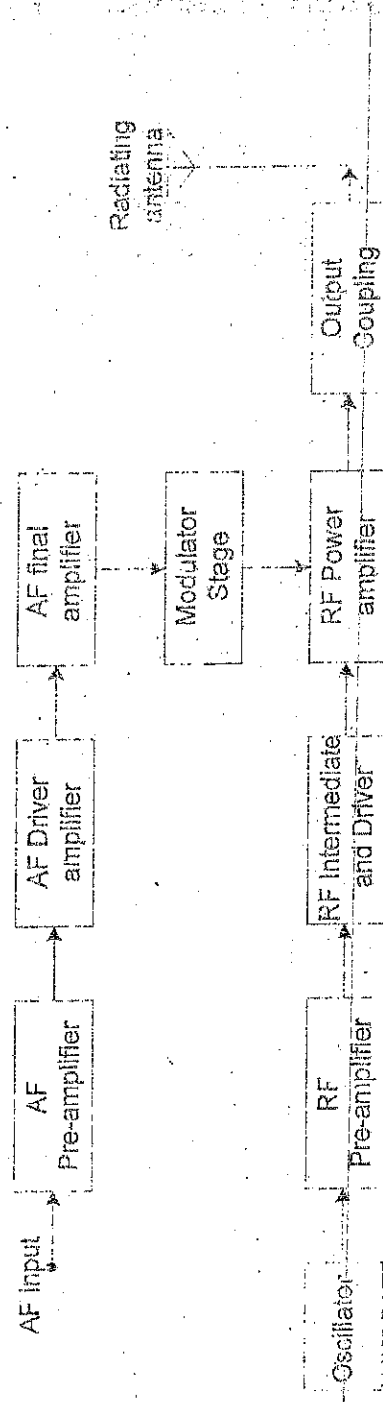


Figure 9: Block Diagram of AM Transmitter

The stages that make up the transmitter are as follows:

(i) Oscillator Stage: Here the carrier frequency is generated. The oscillators at this stage must provide RF voltage precisely at the desired frequency, with a high order of frequency stability, free from hum or noise modulation.

(ii) RF Amplifier: Usually the desired carrier frequency is generated at a low power level by a stable oscillator, ordinarily a crystal oscillator. This is followed by the RF amplifier that increases the power level and isolates the oscillator from the modulator.

(iii) Modulator Stage: For the message signal to be sent out, it must be superimposed upon the carrier signal and transmitted. The process is known as modulation and the stage where this takes place is called the modulator stage.

(iv) AF Amplifier: This set of amplifiers amplify the audio signal before modulation. This is so because the audio signal is of low electrical energy at its input. Thus at its output high energy audio signal is being delivered.

(v) RF Power Amplifier: This further increase the carrier power and the modulated signal power suitable for transmission. Tuned circuits are generally used in the input and output circuits of RF power amplifiers. They are divided into classes A, B and C according to the angle of plate current flow.

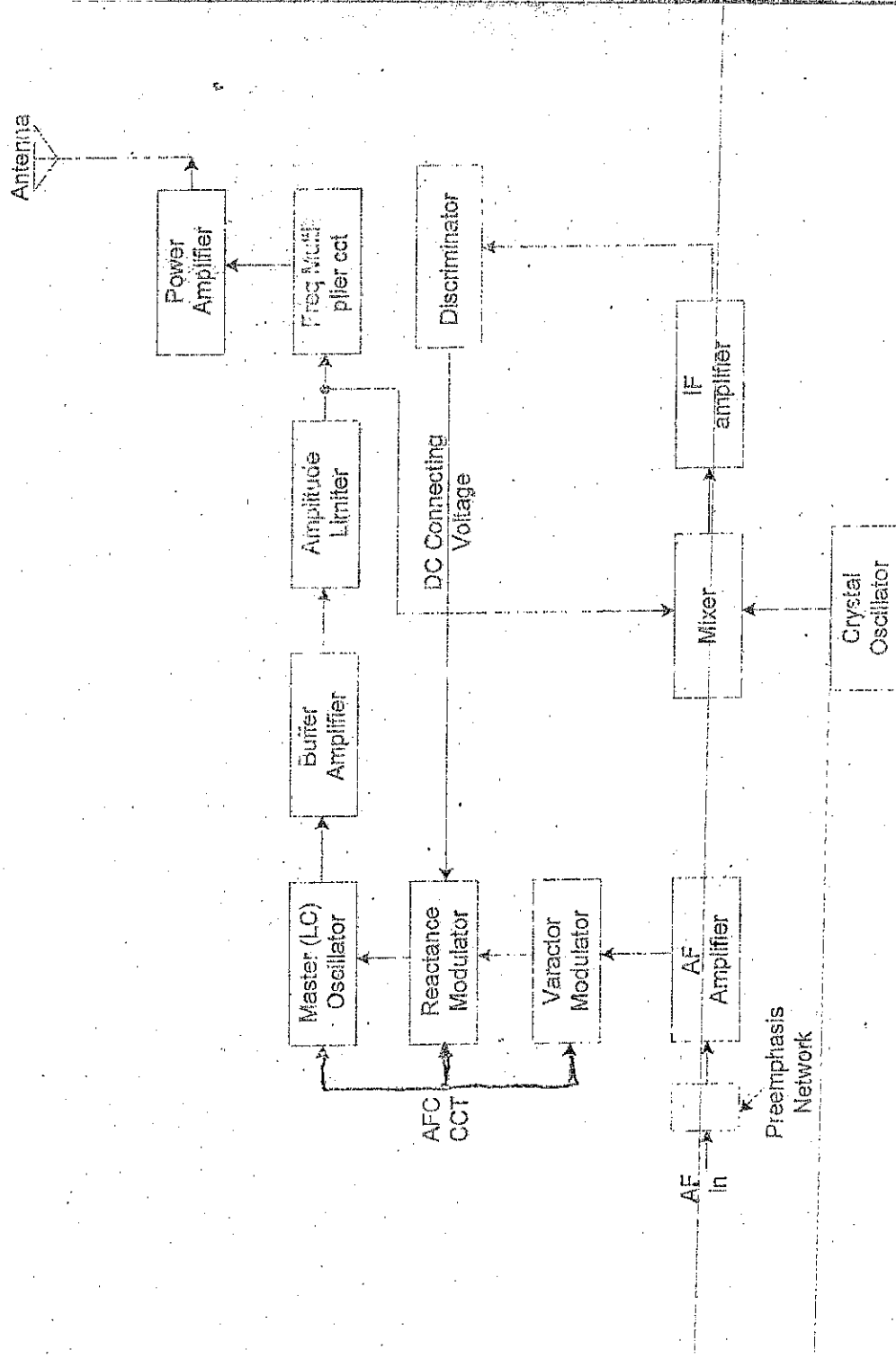


Figure 10: FM Transmitter with AFC Circuit System

Definition of FM:

This is a type of modulation in which the frequency of the carrier is made to vary in accordance with variation in the modulating signal.

Varactor Modulator: This is used to produce the frequency modulation. Most often it is employed together with the reactance modulator to provide automatic frequency correction for the FM transmitter.

Automatic Frequency Control Circuit: This is used to provide frequency stability in the system.

LC Oscillator: This constitutes the master oscillator module. It provides the required tank circuit for the reactance modulator to operate upon and also helps to boost the frequency of the modulated AF signal.

TYPES OF COUPLING FROM POWER AMPLIFIER STAGE TO THE AERIAL

Definition: A coupling network or antenna coupler is a network composed of reactances and transformers, which may be lumped or distributed. The coupling network is sent to provide impedance matching and is employed for any of the following reasons:

(i) To tune out the reactive component of the antenna impedance, to make the impedance look resistive to the transmitter, otherwise detuning will take place when the antenna is connected. This function involves the provision of variable reactances.

(ii) To provide the transmitter (and the transmission line if used) with the correct value of load resistance. This involves having one or more adjustable transformers.

(iii) To prevent the illegal radiation of spurious frequencies from the system as a whole. This function requires the presence of filtering network, generally low-pass, since the spurious frequencies are most likely to be harmonics of the transmitter's frequency.

(iv) For very low frequency transmitters whose output tank is series fed and single-tuned, the antenna coupler must prevent the dc supply from reaching the antenna. This is done to prevent antenna insulation difficulties and danger to operators which ranges from RF burns to a more lethal dc high voltage shock.

The type of coupling system depends on how the antenna is fed, whether a transmission line is used, whether it is balanced or unbalanced and the value of its standing wave ratio.

DIRECTLY FED ANTENNAS:

These antennas are coupled to their transmitters without transmission lines, generally for lack of space. To be of use, a line connecting an antenna to its transmitter ought to be at least a half-wave in length, and at least the first quarter-wave portion of it should come away at right angles to the antenna. This is difficult to accomplish especially at low frequencies, for shipboard transmitters or those on top of buildings.

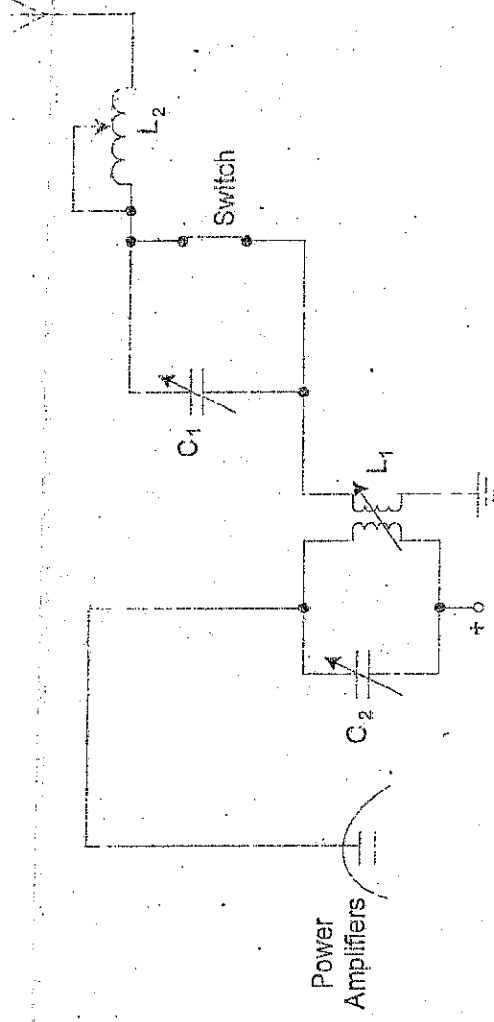


Figure 11: Direct Coupling

The figure above shows the simplest method of direct coupling. The impedance seen by the tank circuit is adjusted by moving coil L_1 or by changing the number of turns with a travelling short circuit. To tune out the antenna reactance, either C_1 or L_1 is shorted out, and the other component is adjusted to suit. This simple coupling network is not the best because it does not attenuate harmonics.

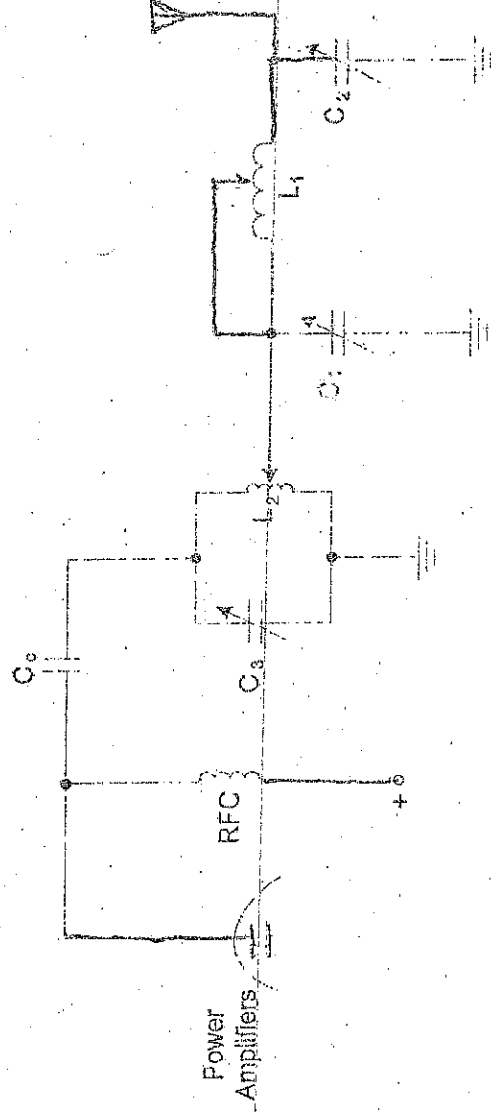


Figure 12: Direct π Coupling

Another method of direct coupling is the use of π - coupling shown in figure 12 above. This is a much better proposition. It affords a wider reactance range and is also a low pass filter, giving adequate harmonic suppression. However, it will not provide satisfactory coupling if the antenna is very short, thus having mostly capacitive input impedance. It is better under these conditions to increase the height of the antenna.

COUPLING WITH A TRANSMISSION LINE:

The requirements are similar to the ones already considered. However, balanced lines and therefore balanced coupling networks are often used; in this case a symmetrical π – coupling network is used as shown in figure 13 below. The output tank

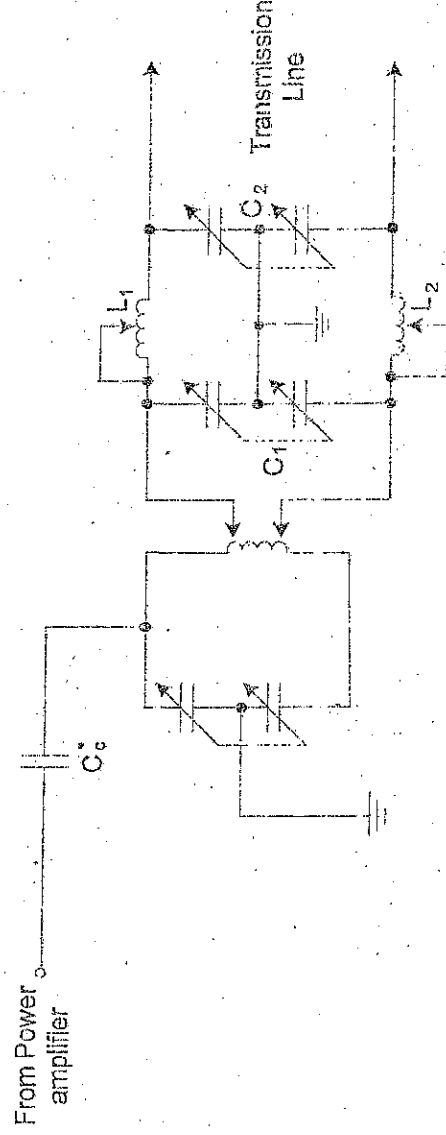


Figure 13: Symmetrical π - Coupler

is tuned accordingly and facilities must be provided to ensure that the two legs of the coupler are kept balanced. At higher frequencies a number of the lumped components can be distributed, so that quarter – wave transformers and stubs can be used.

MEASURING INSTRUMENTS FOR TRANSMITTER TESTING

There are many measuring instruments used in testing transmitters. The most general ones are the ammeters, voltmeters, power meters, frequency meters, oscilloscopes and spectrum analyzers etc.

OSCILLOSCOPES: In the history of electrical and electronic measurement, no instrument has had greater impact than the oscilloscope. Galvanometers and other mechanical apparatus capable of indicating dynamic phenomena existed long before the oscilloscope, but they were machines with slow response. However, the oscilloscope has the advantage of displaying high speed phenomena for easy observation, an asset which grew to greater importance as applications for electronics developed over the years. Today the oscilloscope is the most versatile general purpose electronic measuring instrument available for scientific investigation.

The oscilloscope is an electronic instrument used for display of electrical signals on the vertical axis compared with time on the horizontal axis. Thus the modern

laboratory oscilloscope is a time domain measurement and display instrument for analysis of electrical signals.

The function of the oscilloscope may be defined as capturing, displaying and analyzing a time-domain waveform. Traditionally, the oscilloscope has been used to capture high speed transient phenomena and display them for the engineer or technician to analyze and interpret. The great majority of scopes in use captures a waveform and faithfully portrays that waveform on the CRT for the viewer to observe. Some oscilloscopes today are capable of analyzing data and presenting digital answers for their users. Others in conjunction with computers are capable of wide varying data accumulation and analysis.

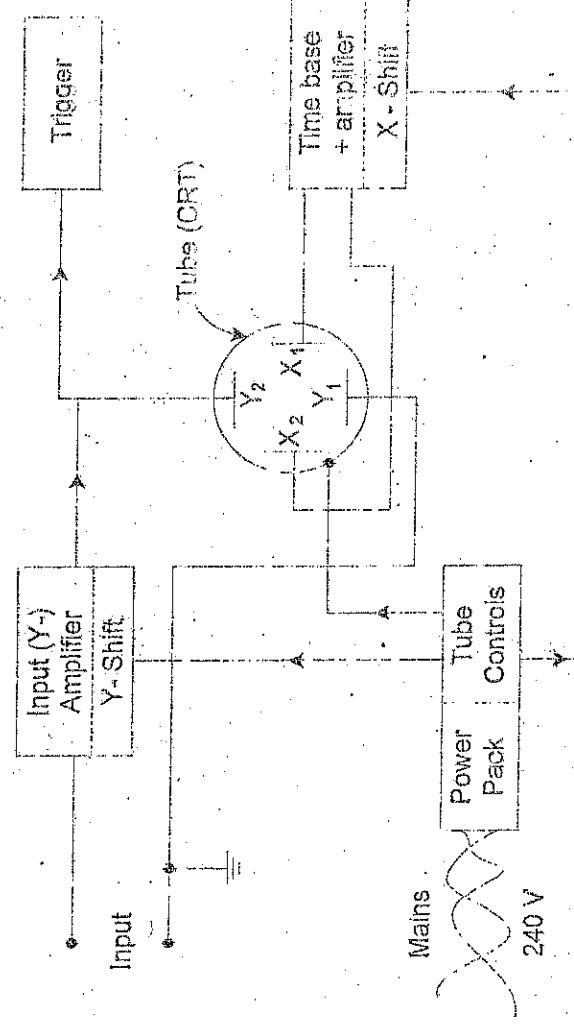


Figure 14: Oscilloscope Block Diagram

The modern time - domain oscilloscope contains the following component blocks:

- (i) Display device (CRT)
- (ii) Power pack supplies
- (iii) Tube Controls: The brightness and focus and correction of astigmatism, are controlled by varying the voltage of the appropriate tube electrode.
- (iv) Y - or input - amplifiers: This is a variable gain linear amplifier which amplifies the signal applied to the Y - plate.
- (v) Y - Shift: This is in the Y - amplifier circuit and it shifts the signal trace up or down on the screen.

- (vi) Time – base: This is a saw-tooth oscillator. In the oscillator circuit, the feedback can be adjusted with stability control until the oscillator is just not free running, and the trigger is then applied to lock the trace.
- (vii) Trigger Unit: This applies a triggering pulse to the time-base oscillator from the Y – or input voltage. The time base frequency is then synchronized (locked) with that of the input signal, so that a stationary trace is obtained on the screen.
- (viii) Time – base amplifier: This amplifies the time base voltage and applies it to the X – plates. The trace can be made to expand or contract horizontally by varying the amplification.
- (ix) X – shift: This shifts the time base horizontally to the left or right.

SPECTRUM ANALYZERS:

A spectrum analyzer visually displays (on a CRT) the amplitude of the components of a wave as a function of frequency. This can be contrasted with an oscilloscope display which shows the amplitude of the total wave (all components) versus time. Thus an oscilloscope shows the time domain while the spectrum analyzer shows the frequency domain.

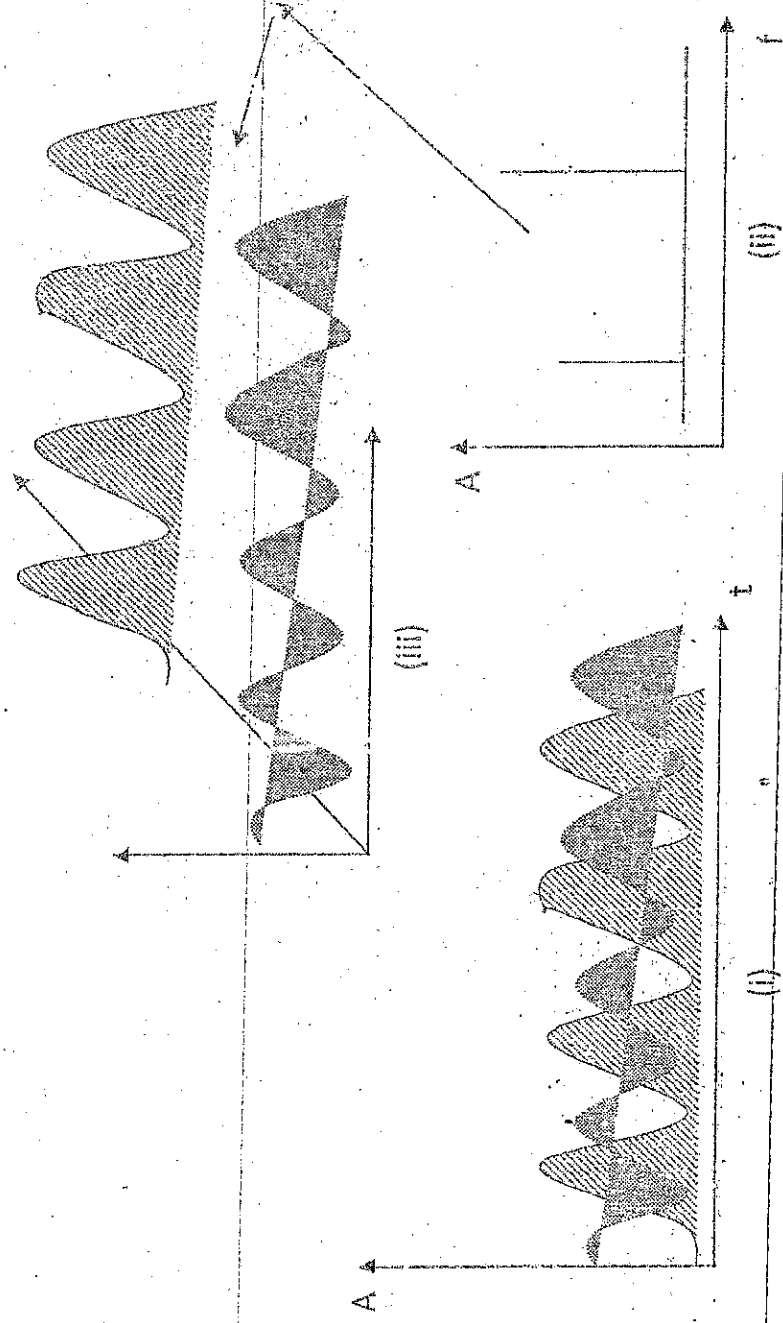


Figure 15: Spectrum Analyzer Principle

Figure 1.5 above shows how a simple signal $A(t)$ looks in the time domain of an oscilloscope (i) and in the frequency domain $A(f)$ on a spectrum analyzer (ii) and (iii).

The frequency domain components are coherent that is, are exact frequency multiples and bear a finite phase relationship to each other. Also several sinusoidal components of differing frequencies that are unrelated to each other are present. These components could be signals from unrelated oscillators, which could be carrier signals of a telephone, a telemetry frequency multiplex system or perhaps several radio stations that are simultaneously present.

Time domain analysis and spectrum analysis of signals complement each other. Spectrum analyzer can measure extremely small amounts of distortion far beyond the capability of an oscilloscope by measuring very small harmonic components generated from a sinusoidal test signal by the device under test.

Spectrum analyzer applications include the measurement of signal level and the measurement of frequency and its response, harmonic and inter-modulation distortion, frequency stability, spectral purity, modulation index and attenuation.